MTHs

YOUNG OPTOMETRISTS

Herpes Simplex Keratitis



[Photo of HSK epithelial disease with dendritic ulcer – NaFl stain]

Signs:

- Corneal epithelial dendritic ulcers with terminal buds (hallmark) – ulcer bed stains with NaFl (edges contain active virus)
- Typically unilateral
- Conjunctival injection, ciliary flush, stromal oedema
- Subepithelial WBC infiltration
- vesicular lesions on eyelids or adnexa
- bulbar follicles
- decreased corneal sensation
- pre-auricular lymphadenopathy
- Oral HSV lesions (cold sores)

Symptoms:

- Painful, red, watery eye
- blurred vision
- photophobia
- usually unilateral.

Diagnosis: Herpes Simplex Keratitis

Differential Diagnoses:

- HZO
- adenoviral conjunctivitis
- CL-related ulcer – MK, CLPU
- Corneal graft rejection
- Keratoconjunctivitis sicca
- Thygeson's superficial punctate keratitis

- neurotrophic keratopathy
- acanthamoeba keratitis
- fungal keratitis
- epithelial trauma
- foreign body.

Risk Factors:
- Infection with HSV (from lesions)
- asymptomatic viral shedding
- reactivated HSV disease (stress, trauma, systemic infection, immunocompromise, menstruation)

HSV characteristics:
- HSV common in most individuals: >95% of individuals > 60yoa
- HSV virus – double stranded DNA virus
- HSV latent in trigeminal ganglion
- HSV-I (ocular) and HSV-II

Aetiology:
- Infectious epithelial keratitis (HSK)
- Neurotrophic Keratopathy
- Necrotising stromal keratitis
- Immune stromal keratitis
- Endotheliitis

Pathophysiology:
- (Re)activation of HSV → infection via skin or moucous membranes → HSV dissemination via mucocutaneous distribution of trigeminal nerve (CN 5 – Ophthalmic division) →

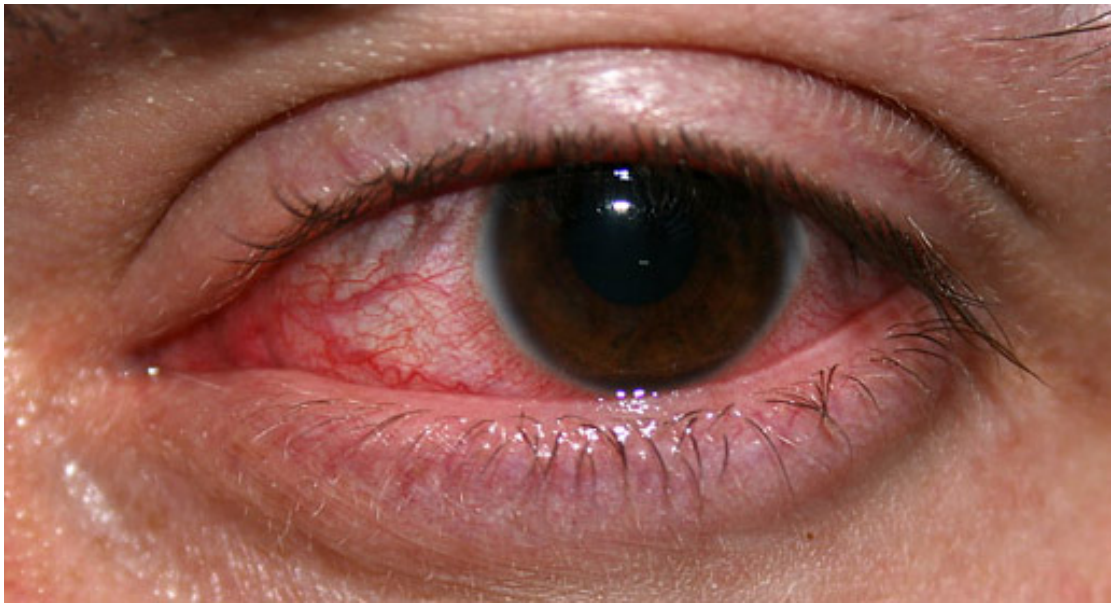
Management:
- Most cases resolve spontaneously within 3 weeks – Tx aimed at minimising corneal scarring and stromal damage
- Epithelial debridement of ulcer edges (active virus) with anaesthetic
- Topical ocular Acyclovir (Zovirax) 3% ophthalmic ointment; 4.5g tube
    - 1cm ribbon, 5 times daily (q3-4h)
    - Review next day
    - Treatment continued for 3 days after resolution
- Acyclovir – synthetic guanosine analogue → inhibits viral DNA polymerase → DNA chain termination → suppresses viral replication
    - Absorbed through corneal epithelial and into aqueous humour
- Oral Acyclovir – 400mg 5 times daily for 7-10 days

Contraindications:
- Known hypersensitivity to acyclovir or valacyclovir
- Steroids contraindicated in acute epithelial disease

    - *** steroids may be used to dampen down inflammatory response once active viral infection has been controlled *** (controversial)
- Pregnancy category B3

Complications:

- Corneal toxicity due to topical anti-virals
- Neurotrophic ulcers
- Geographic (metaherpetic) ulcers
- Disciform keratitis may lead to bullous keratopathy
- Corneal scarring



Background info: 25 year old male, hx of 3 day increasingly red left eye and soreness and sensitivity to light. No discharge, vision is ok. Tried using over the counter eyedrops from the chemist but not any better.

Diagnosis: Unilateral acute anterior uveitis

Signs/Sx:

- Unilateral pain, photophobia, redness and watering developing over one to two days, with little or no effect on vision.
- Pain and photophobia may precede slitlamp signs due to unseen ciliary body inflammation.
- Signs: circum-limbal flush, miotic pupil, anterior chamber cells and flare and small, fine keratic precipitates (KP). The pupil may appear sluggish but is better described as small and irregular.
- In more severe attacks, posterior synechiae, peripheral anterior synechiae, corneal oedema, spill-over vitritis and macular oedema may be seen.
- The IOP generally falls slightly with attacks however, occasionally may increase.

Differential Diagnoses

- Chronic anterior uveitis
- Keratitis
- Conjunctivitis
- Corneal trauma
- Corneal foreign bodies
- Acute angle closure glaucoma
- Scleritis
- Episcleritis
- These are relatively easy to exclude with a slitlamp examination.

Aetiology

There are several factors associated with the cause of AAU:

- Association with HLA-B27
- Idiopathic (HLA-B27 negative)

- Other systemic disease (related to infection or inflammation)
- Cases that are associated with HLA-B27, idiopathic, secondary to herpetic keratitis are relatively common. Other conditions are relatively rare
    - A large case of cases are idiopathic. Stress is often considered to affect immunity and AAU might be triggered by stress.
- Ask about the patient's general health. A review of systems can be considered for conditions known to be associated with AAU. These include
    - Rheumatological: spondylarthropathies, esp ankylosing spondylitis (e.g. ask if any joint pain, back pain)
    - Genito-uritary: Reiter's syndrome, Behcet's, syphilis
    - Respiratory: Sarcoidosis, tuberculosis (persistent cough?)
    - Gastrointestinal: inflammatory bowel disease (diarrhea, blood in stools?)
    - Dermatological: psoriasis, syphilis, sarcoidosis, herpes simplex, zoster, Behcet's (ulcers, rashes?)

Pathophysiology
The clinical picture seen in AAU is intimately related to the breakdown of the blood aqueous barrier. This results in inflammation, with the typical groups of signs and sx. Many AAU signs are seen only because of changes in the permeability of the iris and ciliary body capillaries. These result in an escape of leukocytes, protein and fibrin into the aqueous with the potential for structural side effects such as synechial adhesion to the lens and anterior chamber wall.

Management
Key goal is to control the inflammation, which includes aggressive use of topical corticosteroids. In general, the greates anti-inflammatory effect is found with prednisolone acetate, fluorometholone acetate and dexamethasone.
Hourly instillation of steoids for the first few days is typical. A loading dose to achieve therapeutic concentration at an early stage can be recommended e.g. one drop every 15 mins for the first hour, then every 30 mins for the first two hours. To continue this aggressive early approach, consideration should be given to night time schedule of drops too. More severe cases may require hourly or two hourly instillation throughout the night. An alternative may be to double the dose during waking hours or to use cortisone ointment at night.

The need to shake bottles should be stressed, as well as allowing enough time in between instilling different medications.

Patients should be reviewed soon after tx is initiated. Typically, the first review will be the next day. The signs of inflammation may have only changed minimally at this time. Continue to monitor and tapering of the steroid drops can be initiated as improvements are seen.

Cycloplegia should be considered to reduce in pain and photophobia by reducing iris spasm and ciliary body function. Symptomatic relief is usually prompt and reported at review. This also acts to break any existing PS and preventing the development of new PS. Cylcoplegia may also assist in stabilizing the blood aqueous barrier.

Frequent dosage such as four times a day of homatropine may be needed. Cyclopentolate may act as a chemo-attractant for leukocytes and should be avoided in the presence of inflammation. In patients with mild inflame and no synechiae at initial presentation and no synechiae at presentation, some clinicians may feel tropicamide is effective enough. For those with very mild inflammation and no synechiae at

presentation they may be managed without mydriasis (other than that required at their first visit for proper examination i.e. to rule out vitreous cells and cystoid macular oedema).

Potential complications

- A significant portion of patients are steroid responders who may develop a IOP rise over a period of days to weeks. It is not feasible to withdraw steroid treatment. If the iOP is high enough to sufficiently cause damange to the optic nerve or if the px already has ON damage, topical meds to lower IOP may be used. For example, timolol, brimonidine or CAI can be considered. Prostaglandin analogues should be avoided.
- In the short term, small pressure rises may only need observation.

Referral

- Unsure of diagnosis
- Not responding to tx
- Consider referring with the presence of severe or atypical inflammation
- Hypopyon
- Bilateral presentation
- Recent surgery or a presence of a drainage bleb

The Case

A 27 year old male presented with a red and watery left eye for 10 days duration. He complained of some photophobia to his left eye and had noticed yellow crusting along his eyelashes in the morning and white discharge. He had previously obtained chloramphenicol 0.5% (Chlorsig) eye drops and had been using them for the last 7 days in his left eye qid with no improvement to symptoms. His right eye was normal. Otherwise he was feeling healthy, in good health and taking no medications.

What tests would you do?

- Vision
- Pupils
- Slit lamp examination - including NaFl, everting eye lids
- IOPs
- Adeno-detector ?
- Swabs ?

Examination findings are as follows:

Vision was 6/6- right and 6/7.5- left.

SLE:
RE - normal
LE - follicles of the superior and inferior tarsal plates.
conjunctiva diffusely hyperaemic although noticeably worse superiorly, small nodule of episcleritis was apparent.
cornea - clear
anterior chamber quiet

Any other tests you would do now?

- Refer for swabs (always swab non-responsive cases) and PCR testing,
- Adeno detector
- In room povidine iodine lavage
- Check for tender lymphadenopathy.

- Gram staining, ADV PCR, HSV VZV PCR- nothing detected. No growth after 48 hrs.

Why are you thinking now as your initial diagnosis?

Chlamydial conjunctivitis - Age of patient (young adult), unreactive to AB treatment and non self limiting, follicular conjunctivitis with mucopurlent discharge. Bacterial typically has tarsal papillae and no-pre auricular lymph node involvement (except when systemic cause).

Symptoms - foreign body sensation, mucopurulent discharge and ocular irritation.

And differential diagnosis — Molluscum contagious, **allergic/toxic conjunctivitis** from excessive AB use? some individuals have follicles normally. Young adult and presentation and non-responsive to topical AB suggest chlamydial conjunctivitis.

Adult chlamydial conjunctivitis is typically acute/subacute in onset, unilateral and presents with mucopurulent discharge. Follicles most prominent in the inferior forniceal conjunctiva is **characteristic** but may also involve the upper tarsal conjunctiva.

Other possible diagnoses - **Adenoviral conjunctivitis** (watery discharge most common, photophobia), acute follicular conjunctivitis - inferior conjunctiva more severely affected. Unilateral or bilateral , one eye followed days later by the other eye. Symptoms get worse for 7-10 days. EKC - typically bilateral and has a stream keratitis in 80% of cases (non corneal involvement in this case - diffuse epithelial punctuate keratitis and tiny marginal infiltrates after several days to one week, sub-epithelial infiltrates appear at 2 weeks and may persist for weeks), no systematic involvement (true for this case). and PCF - patient would have a fever, pharyngitis - most common in Children. Redness worst usually inferiorly NOT superiorly as in this case. Seri-Fibrinous discharge - not mucopurlent discharge as in this case. Ocular symptoms commonly preceded by system symptoms of fever, sore throat and sometimes gastrointestinal disturbances in PCF (non of which is present in this case).

HSKconjunctivis - lid blisters, unilateral, cold sores, past history and poor corneal sensitivity,

Contact lens infiltrative keratitis

PCR testing will differentiate HSVK, Adenoviral and Chlamydial conjunctivitis. Location of follicles and infiltrates can aid diagnosis.

Trachoma ? Also caused by *Chlamdia trachomatis*. Chronic inflammatory follicular conjunctivitis. A disease of poverty - occurs in areas with poor personal and community hygiene. Most commonly in children aged 3-5 years.

Other clinical signs of chlamydial conjunctivitis include chemosis, peripheral corneal infiltrates 2-3 weeks after the onset of the conjunctivitis and tender lymphadenopathy. The patient in this case report had noticed his eyelids were swollen. Stenburg and Mardh found 17 of their 60 adult subjects with Chlamydial conjunctivitis had swollen eyelids because of inflammatory oedema in the upper eyelid. [4]

Management?

- Referral to general practitioner for further investigations and treatment is required. And for sexual partners.
- Patients should be re-tested after 3 months.
- Diagnosis of chlamydial conjunctivitis is confirmed by laboratory investigations including swab and culture, most commonly Giemsa staining and polymerase chain reaction (PCR).
- Topical treatments are ineffective in eradicating the eye condition.
- Systemic therapy is most commonly oral azithromycin (zithromax) 1g as a single dose.
- Alternative treatments: erythromycin 500 mg qid for 1 to 3 weeks or doxycycline 100mg bd for 10-14 days and are equally effective.
- Erythromycin safe in Pregnancy (class A).
- Azithromycin is category B. Pregnant women with an untreated chlamydia infection are at a **greater risk** of developing complications such as a miscarriage, premature birth or stillbirth.
- Supporting the eye with non-preserved lubricants and saline baths can help improve ocular symptoms.

Disease aetiology?

- Small, gram negative bacteria that depend on host cell for energy. Resemble virus.
- Most common bacterial sexually transmitted disease
- Serotypes D trough K produce adult inclusion conjunctivitis, Serotypes A to C cause Trachoma
- Transmission to eye can occur by: hand contact, a mother to new born or towels, swimming pools, spa baths.
- Swab shows intracytoplasmic inclusion bodies in epithelial cells, neutrophils and lymphocytes.

Young Optometrists
Mock Oral Therapeutics Examinations (MOTHs)

Pseudoexfoliation (PXF) Glaucoma

Clinical Scenario

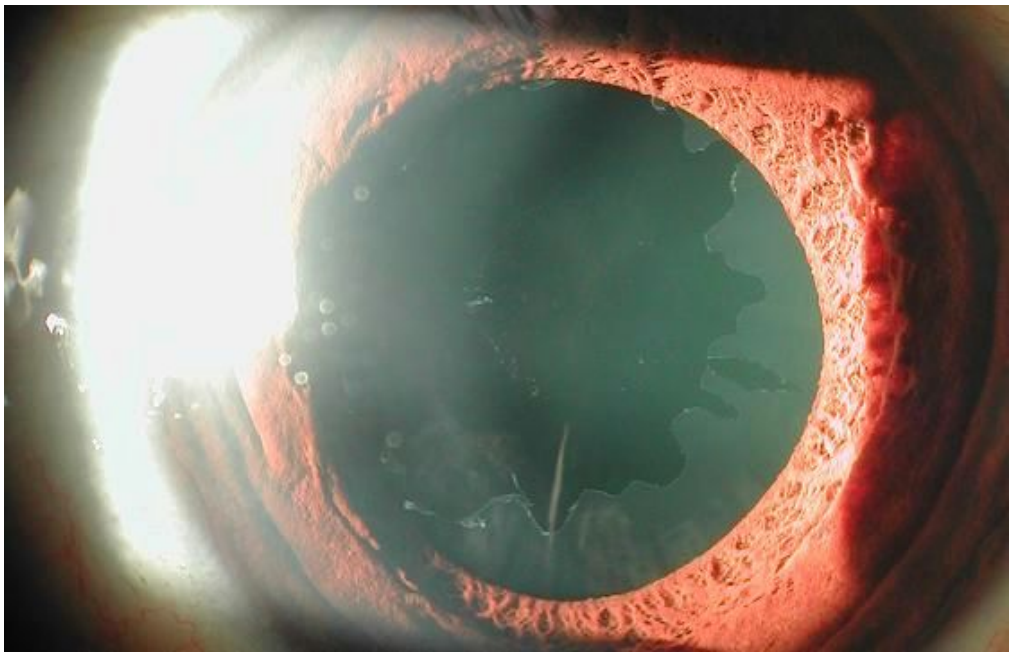
A 65-year-old Caucasian man presents to you reporting mild blur for the past year. Past ocular history is unremarkable. He is currently taking Lipitor for high cholesterol.

His VA is 6/9 in each eye. Applanation tonometry is R 23mmHg and L 15mmHg at 10am.

The patient is dilated but has poor pupil dilation.

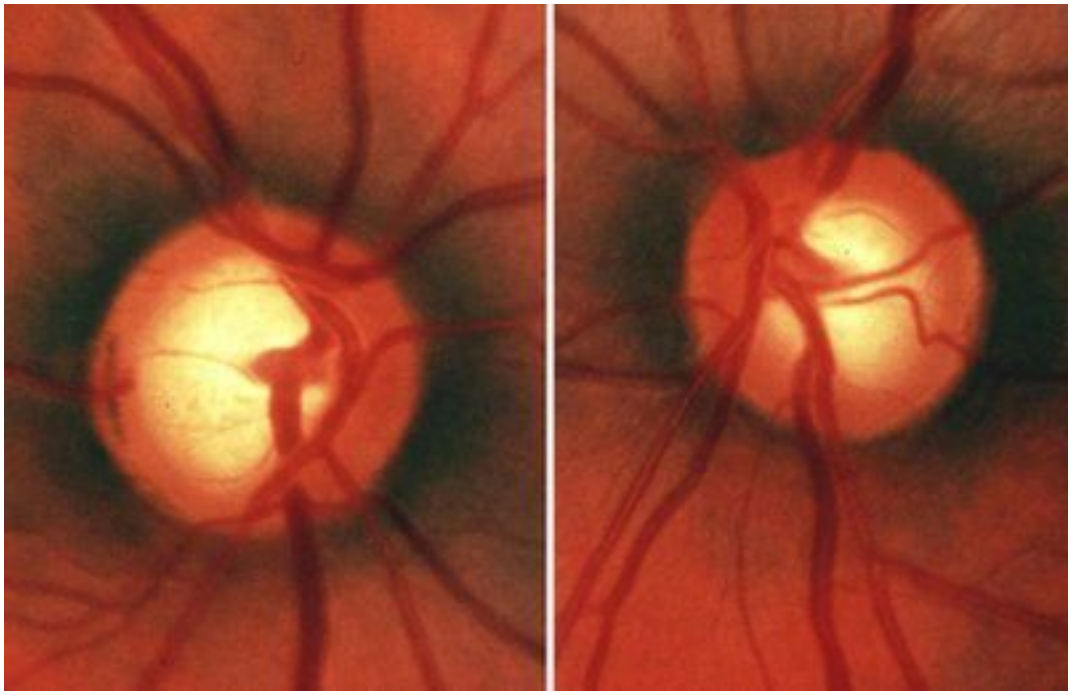
Please describe the anterior features seen in the image below.

There is white pseudoexfoliative material on the pupil margin and the anterior lens surface with a clear intermediate zone.



Please describe the optic discs below

Asym CDR R>L correlates with asym IOP; Vert CDR 0.8/0.6; average ONH size OU,
OD Baring of superior and inferior b.v with associated NRR thinning, laminar pore show
OS Baring of superior b.v., inferior NRR thinning



What is the most likely diagnosis and what are the differentials

Most Likely Diagnosis: ***Pseudoexfoliation glaucoma***
DDx: ***Primary open-angle glaucoma, Primary angle-closure glaucoma, Pigment dispersion syndrome (PDS), true exfoliation (found in glass blowers aka capsular delamination refers to a separation of the superficial layers of the lens capsule due to heat or infrared radiation), amyloidosis***

Work up: what other tests do you want to do?

1. *Gonioscopy*
2. *Pachymetry*
3. *OCT*
4. *Humphrey Visual Field Threshold 24-2*
5. *Iris transillumination defects*
6. *Ask about FHx of glaucoma*

What would you expect on gonioscopy?

Uneven TM pigment distribution, Sampaolesi's line (pigment anterior to Schwalbe's line)
White pseudoexfoliative material deposited in angle

What would you expect on gonioscopy?

Patient age: *older age in PXF, young for PDS*
Asymmetry: *usually more asymmetry in PXF, more bilateral in PDX*

Amount and location of pigment: *PDS has more pigment, PXF has white pseudoexfoliative material, uneven distribution in PXF*
Location of transillumination defects: *mid-peripheral transillumination defects in PDS, around the pupil in PXF*
Krukenberg spindle: *classic in pigmentary glaucoma, much less often seen in PXF*

Why do you get poorly dilating pupils in PXF?

This may be the result of structural damage to the iris caused by the pseudoexfoliative debris.

Why is there a clear intermediate zone in PXF on the anterior lens surface?

The pseudoexfoliative material is rubbed off by the iris, leading to the clear intermediate zone. Most easily seen after dilation. Note, this material may also be seen on the conjunctiva, corneal endothelium, lens zonules, ciliary processes and trabecular meshwork.

Risk factors for PXF

Older age usually late 60s-70s
Scandinavian and Mediterranean race

Management: how would you manage this patient?

Referral to glaucoma specialist (ophthal)

Medical therapy first – 30% target pressure

High IOP fluctuations so it challenging to reach the target pressure, need 24-hour IOP control with PGA

PGA Options:
Latanoprost, Xalatan 0.005%
Travoprost, Travatan 0.004%
Bimatoprost, Lumigan 0.03%, available in preservative free
Tafluprost, Saflutan 15 mcg/ml, preservative free
e.g. Lumigan PF nocte OU, Educate on proper drug instillation

Referral to GP regarding systemic health as exfoliative material can deposit on heart, lungs and kidney

Frequent review initially to establish progression with IOP, gonio, disc evaluation, OCT, VF every 3 months, then extend to 6 monthly (highly progressive)

What are the contraindications for PGA in general?

Pregnancy Category C

Known hypersensitivity

Relative CI: uveitis, cystoid macular edema, HSK

What are the side-effects of PGA?

Iris darker, conjunctival hyperaemia, eyelash hypertrichosis, PAP, i.e. periorbital fat atrophy, mild enophthalmos, eyes look sunk in

Prognosis

According to Early Manifest Glaucoma Trial (EMGT), worse prognosis for PXF compared to POAG with faster rates of progression (two-fold increase) and poorer response to medications and increased need for surgery

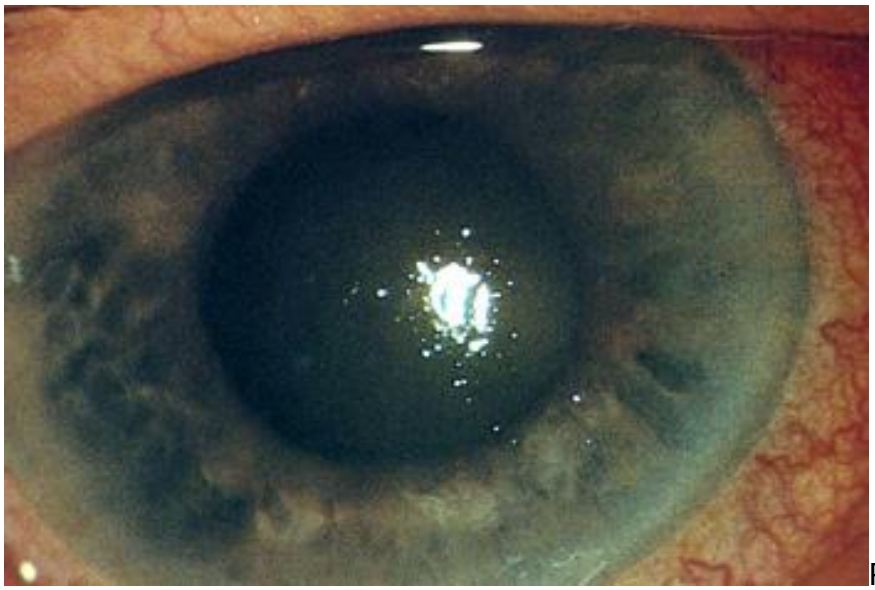
Discuss the advantages and disadvantages of cataract surgery for this patient

Advantages: *Rx corrected, cataract out so VA better, may reduce IOP*
Disadvantages: *Weak zonules so risk of lens subluxation, poor dilation, steroid response higher in glaucoma, higher IOP during cataract surgery*

Describe the pathogenesis of PXF glaucoma

Fibrillar elastotic material is deposited in trabecular meshwork, on anterior lens surface, zonules, pupillary margin → alters metabolism and disturbs structure and function of iris → material clogs up TM so reduced aqueous outflow and rasied IOP
Constant pigment liberation causes defect at pupillary margin of iris from rubbing of iris against rough lens capsule → releases more pigment hence the pigmented TM, Sampaolesi's line which blocks TM to reduce outflow → glaucoma

YO – MOTHs Orals


Photo supplied with NaFl stain

Diagnoses: Acute Angle Closure

Symptoms

- Hazy or blurred vision
- The appearance of rainbow-coloured circles around bright lights
- **Severe eye and head pain
- **Nausea or vomiting (accompanying severe eye pain)
- Sudden sight loss
- Redness
- Tearing
- Photophobia
- **History of previous attacks

Signs:

- Middle aged / elderly patient, Female > Males
- Pupil is mid-dilated and non-reactive
- Cornea oedematous (clear reflex is absent)
- Eyeball feels "hard"
- IOPs greater than 21mmHg
- Conjunctival injection

REASONABLE Differential Diagnoses (considering signs/symptoms):

- Anterior Uveitis / Iritis / Uveitis
- Endophthalmitis

Pathophysiology:

- Egress of aqueous humor circulation is compromised and IOP is subsequently increased
- Primary versus secondary causes
- Aqueous humor is produced by ciliary body in the posterior chamber
    - Diffuses from the posterior chamber, through the pupil, and into the AC
        - From AC the fluid is drained in the vascular system via the trabecular meshwork and Schlemm's canal
- Dilation (in response to environmental or chemical stimuli results in a pathologic iris-lens apposition). The contact and apposition between the lens and the iris is called:
    - ***Pupillary block***
        - The forward-most surface of the lens is anterior to the plane of the iris insertion into the ciliary body.
        - Aqueous flow from PC to the AC is obstructed or blocked.
        - Increasing pressure in the posterior chamber causes the pliable iris to bow forward leading to **iris bombé**
            - Further closes the already narrow angle, and compromises aqueous drainage, increasing IOP
- Research suggested an alternative pathophysiologic pathway (Cronemberger et al)
    - Acute events can be traced to an autonomic imbalance, specifically increased sympathetic tone in individuals with AACG.
    - Iris dilator muscles have been found to be more developed and stronger
        - In instances of increased ocular sympathetic tone
            - Emotional distress
            - Low light conditions
            - Sympathomimetic drug use
        - Contraction of iris dilator muscles leads to pupil dilatation and thickening of middle-peripheral iris. This thickening can lead to angle closure, obstructing outflow of aqueous humor.
- Other mechanisms include (non-exhaustive)
    - plateau iris – anterior insertion of iris, crowded iris tissue blocks trabecular meshwork, causing increasd IOP
    - lens swelling – cataracts (hydration forces cause enlargement of lens, crowding the AC. Forces posterior to the lens can also push the lens and iris forward causing ciliary block / vitreous pressure)
    - ciliary block – when the aqueous is misdirected to go into the vitreous body. Most commonly after surgeries

Aetiology:

- Anatomic abnormalities, lead to AC crowding
    - Shallower AC chambers, thinner ciliary body, thinner iris, anteriorly situated thicker lens, shorter axial eye length
- Narrow angle
- Drugs (sympathomimetics, anticholinergics, antidepressants (SSRIs, anticonvulsants, sulfonamides, cocaine, botulinum toxin)
- Dim light, rapid correction of hyperglycemia

Treatment:

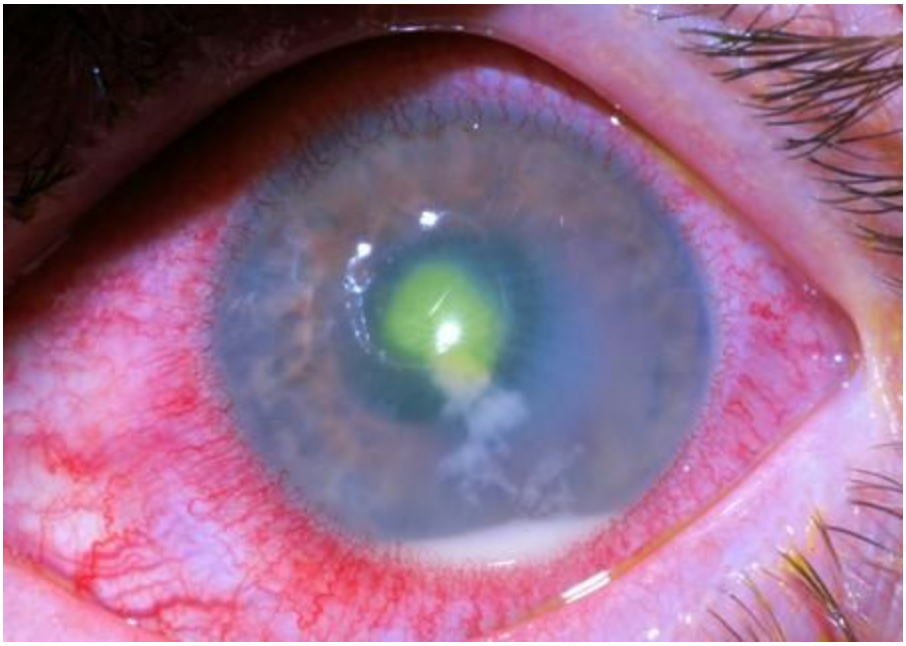
- Patient Outcome aimed at:
  - IOP reduction
  - Suppressing inflammation
  - Reversal of angle closure
- From Optometry board website: in urgent cases, a pharmacist can supply Acetazolamide (Diamox) to the patient when requested to do so via telephone order from a medical practitioner.
  - Optometrists are advised to liaise with a medical practitioner to arrange for the supply of Diamox
    - Oral dose of acetazolamide 500mg (Diamox) can be administered provided there are no contraindications
      - If Diamox cannot be treated orally, patient is likely to require intramuscular or IV administration.
      - In this case referral and transfer to ophthalmologist or hospital emergency department should occur without delay
    - Instill following topical medications (approximately 2 mins between each drop)
      - 1 drop beta blocking agent (e.g. timilol 0.5%)
      - 1 drop alpha agonist (e.g. brimonidine 0.2%)
      - 1 drop carbonic anhydrase inhibitor (e.g. brinzolamide 1%)
    - If eye is red and inflamed:
      - 1 drop high penetrant topical steroid (prednisolone acetate 1%) – with the optometrist or patient to instill a further 3 drops regular intervals in first hour than hourly thereafter.
    - Preferable to withhold pilocarpine instillation until IOP has been reduced
      - EXCEPT in rare instance of angle closure event being caused by routine pupil dilation in which instillation of pilocarpine should commence immediately.
    - Just before patient is transferred and after discussion with the ophthalmologist confirming that instillation of pilocarpine is indicated, instill
      - 1 drop pilocarpine 2% - with the optometrist or patient to instill a further drop in 15 mins)
  - NB// Pilocarpine should only be used for mechanisms involving phakic pupillary block or angle crowding; use of pilocarpine is contraindicated for retrolenticular causes of angle closure as well as those involving aphakic or pseudophakic closure events.

Other treatments / Contraindications:

- Addressing extraocular manifestations is also critical
    - Includes analgesic for pain and antiemetics for nausea and vomiting, can drastically increase IOPs
- Place patient in supine position – aid comfort and reduce IOP. Lens could fall away from iris and decrease pupillary block
- Pilocarpine **–** Why do you wait for IOP to be reduced?
    - In initial attack the elevated IOP in the AC causes a pressure-induced ischemic paralysis of the iris >> Pilocarpine would be ineffective
    - Concerns about its mechanism of action: Constricts ciliary muscle, increase axial thickness of lens and induces anterior lens movement
        - Could result in reducing depth of AC and worsening clinical situation
- Choong et al identified satisfactory reduction as IOP less than 35 mmHg or reduction greater than 25% of presenting IOP.
- If IOP is not reduced 30mins after second dose of pilocarpine, an osmotic agent is considered
    - Glycerol can be administered in non-diabetics
    - Oral isosorbide is administered in diabetics (to avoid the risk of hyperglycaemia)
    - Patients unable to tolerate oral intake or do not experience decrease in IOP are candidates for IV mannitol
- Hyperosmotic agents:
    - Reduce vitreous volume, decreasing IOP and reverses iris ischemia
    - Improves responsiveness to pilocarpine and other drugs
    - Osmotic diuresis and total body fluid reduction
        - Should NOT be administered in cardiovascular and renal patiets
    - Choong et al demonstrated 44% patients required the addition of an osmotic agent
- Corneal Indentation (when medical therapy proves to be ineffective)
    - Temporary measure to reduce IOP until definitive treatment is available.
    - Cornea is indented, AH is displaced to the periphery of anterior chamber which serves to temporarily open the angle
    - IOP is reduced and occasionally may completely abort the attact
    - Apply topical anaesthetic, and smooth instrument can be used (gonioprism, cotton tipped applicator
        - Concern with corneal epithelium
- Laser peripheral iridotomy performed 24-48 hrs after IOP is controlled is the definitive treatment for AACG. LPI may also be prophylactically to individuals anatomically predisposed to AACG if identified before first attack.
- Argon laser peripheral iridoplasty – burns are made in the peripheral iris resulting in iris contraction and opening of angle.
- Lens extraction, proven to effectively reduce IOP without the need for medication post-op.
    - Good for co-existing cataracts
- Ocular massage through a closed eyelid may be preformed while waiting for ophthalmology if no other treatment reduces IOP.
- If the IOP is unchanged or increased from the time of treatment, further treatment should be discontinued and the attack most likely will terminate only with LPI

Must always refer or co-manage with ophthalmologist.

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Photo supplied with NaFl stain

Diagnoses: Contact Lens Related Bacterial Keratitis

Symptoms

- Sudden onset of pain, rapid progress (1-2 days)
- Redness
- Tearing
- Photophobia
- Blurred vision

Signs:

- Epithelial defect
- With stromal involvement (greater than 1mm)with indistinct edges (suppurative stromal infiltrate). The infiltrate may be associated with stromal necrosis and progressive thinning and/or perforation
- Edema and white cell infiltration in surrounding stroma (there is often a characteristic, “ground glass” appearance to the noninvolved areas of the cornea adjacent to the active microbial process)
- Inflammatory endothelial plaque
- Hypopyon (anterior chamber reaction often seen), look for cell and flare, fibrin, hyphema

- Conjunctival injection
- Contralateral eye for clues to etiology as well as possible similar underlying pathology.

REASONABLE Differential Diagnoses (considering signs/symptoms):

- Acanthemoeba Keratitis
- Fungal Keratitis
- Endophalmitis from bacterial keratitis

Pathophysiology:

- Intact corneal epithelium becomes interrupted and/or abnormal tear film allows entrance of microorganisms into the corneal stroma.
- Microorganisms can proliferate and cause ulceration
- (High virulence may initiate microbial invasion – bacteria displaying several adhesins on fimbriated and nonfimbriated structures that may aid adherence to host corneal cells.)
- Initially epithelium and stroma in area of injury and infection swell and undergo necrosis
- Acute inflammatory cells (neutrophils) surround the beginning ulcer and cause necrosis of the stromal lamellae
- Diffusion of inflammatory products (cytokines) towards AC elicits an outpouring of inflammatory cells into the AC and may create hypopyon.
- Bacterial toxins and enzymes (elastase / alkaline protease) may be produced during infection and contribute to destruction of cornea.
- Most common bacteria: Streptococcus, Pseudomonas, Enterobacteriaceae and Staphlococcus.

Although a variety of bacterial and fungal pathogens may cause SCL-related MK, most cases are caused by *Pseudomonas aeruginosa* (9). Bacterial keratitis caused by this Gram-negative organism is associated with a worse visual prognosis than that caused by most other common bacterial pathogens

Aetiology:

- Overwearing contact lenses

- Lack of hygiene increases prevalence of bacteria on contact lenses and contact time on the eye. (including not disinfecting, not cleaning cases, storing in water, topping off solution, sharing lenses)
- Recent eye injury
- Corneal disease (herpetic keratitis, neurotropic keratopathy)
- Weakened immune system (diabetes
- Problems with eyelid (malposition) / tearing (aqueous tear deficiency)

Treatment:

- Patient Outcome aimed at:
    - Reducing pain
    - Resolving discharge as well as corneal and anterior chamber inflammation
    - Resolving epithelial defect
    - Restoring corneal integrity and minimzing scarring and vascularization and vascularization
    - Restoring visual function
- Topical braod-spectrum antibiotics are used initially e.g. Ciprofloxacin 0.3% and Ofloxacin 0.3%
    - Central or severe keratitis (deep stromal involvement or infiltrate >2mm with extensive suppuration)
        - Loading dose is required (e.g. every 5-15 min for the first 30-60min)
        - Followed by frequent applications (e.g. every 30min-1hr for 1-2 days)
    - Less sever keratitis, a regiment with less frequent dosing is appropriate
- Cycloplegic agents may be used to decrease synchia formation and to decrease pain in bacterial keratitis and are indicated when substantial anterior chamber inflammation is present.
- Fluoroquinolones are active against both gram-negative and gram-positive bacteria (Staph aureus).
    - Inhibits DNA gyrase, a type II topoisomerase, and topoisomerase IV, which are enzymes necessary to separate bacterial DNA, inhibiting cell division.

Contraindications / Costicosteroid Therapy

- Some may have a beneficial role in treating some cases of infectious keratitis by:
    - Supressing inflammation
        - Reduce subsequent corneal scarring and associated visual loss
- Disadvantages
    - Local immunosuppression
    - Infection can return
    - Inhibits collagen synthesis, predisposing corneal melting
    - Increase in IOP
- Studies both argue that there is no benefit with concurrent topical corticosteroid therapy in conjunction with a broad spectrum antibiotic
    - but also there is also no increase in adverse events with corticosteroid use.
- Despite risks, many experts believe that corticosteroids can reduce morbidity
    - BUT patients being treated at the time of presentation *should not be on corticosteroids until the infection has been controlled*.
- Inflammation and symptoms (vision, photophobia, lacrimation, injection, and hyperemia) may temporarily increase as corticosteroids are reduced due to lack of inhibiting the local immune response.
- If practitioner chooses to use topical corticosteroid therapy:
    - Should use minimum amount required to achieve control of inflammation
    - ***Requires optimal timing, careful dose regimen***
        - ***CONCOMITANT antibacterial medication and close follow up***
- A conservative approach would avoid prescribing corticosteroid treatment for presumed bacterial ulcers until the organism has been identified.
    - If ulcer is associated with Acanthemoeba or fungus or HSV, outcomes of steroid therapy are likely poor
- For most other bacteria the risk is low and in some cases may be beneficial
- ***If corneal infiltrate compromises the pupil, corticosteroid therapy may be added following at least 2-3 days of progressive improvement with topical antibiotic treatment*** (after fungal infection has been ruled out).

- Topical antibiotics, which are generally administered more frequently than corticosteroids during treatment of active infection are continued at high levels and tapered gradually
- Patient compliance is essential, IOP must be monitored
- The patient should be examined 1-2 days after initiation of topical corticosteroid therapy.

In daily practice, any suspected bacterial keratitis >2mm, or that could compromise vision should be referred to a corneal specialist for danger of losing vision.

MOTHs

YO YOUNG OPTOMETRISTS

Adenoviral keratoconjunctivitis

Case History:

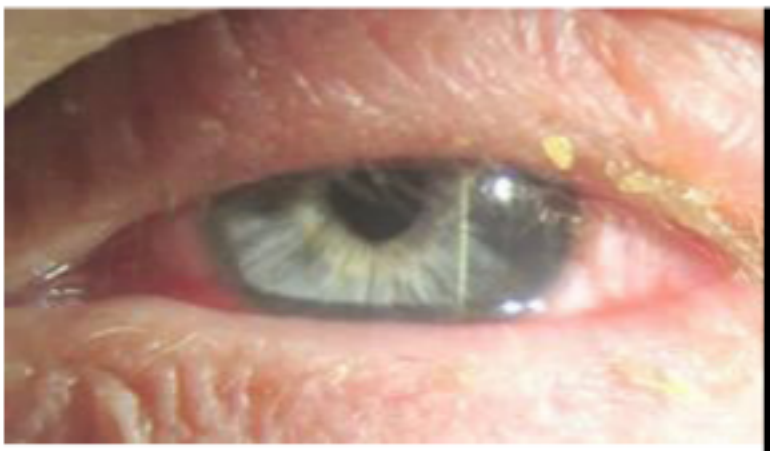
- 25-year old white female
- presented with a 5 day history of watery, red and irritated eyes, which is worse in the left
- Started in left eye first, started with slight soreness, foreign body sensation with mucous discharge that progressively worsened and involvement of right eye 2 days later. Lids feeling stuck together upon wakening.
- There was no reduction in vision, photophobia or itch.
- She reported a short-term alleviation of her symptoms with non-preserved Refresh lubricants.
- **Ocular history:** No previous ocular problems. No previous eye surgery, trauma, or contact lens use.
- **Medical History:** no systemic conditions, generally healthy. No medications or allergies.
    - One week prior to her ocular symptoms starting, the patient had an upper respiratory tract infection.

Examination:

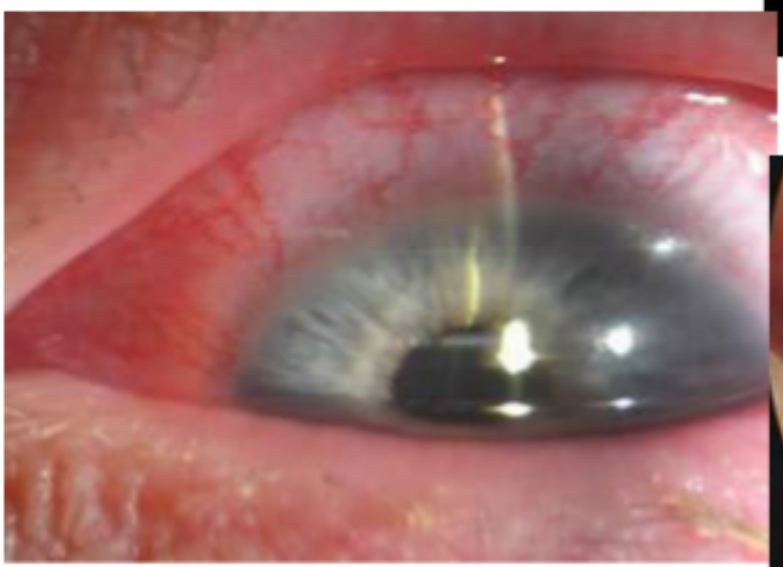
- Vision was R 6/6- L 6/7.5-
- Ocular motilities were full and extensive and confrontational perimetry found full fields OU.
- Pupil reactions were normal with no relative afferent defect.
- Slit-lamp examination found:
    - The eyelids were hyperaemic and oedematous, which was worse in the left eye.
    - crusting of the lashes OU

- grade 3 diffuse bulbar conjunctival hyperaemia in the left eye, compared to grade 2 in the left. The left palpebral conjunctiva had diffuse grade 3 redness while the right eye had grade 2 hyperaemia.
- Follicles were present on the inferior palpebral conjunctiva OU.
- The right eye was watery with a tear prism of 0.5mm compared to 0.3mm in left.
- No corneal infiltrates were detected, but there was inferior SPK which was L>R. No cells or flare were detected in the anterior chambers.
- Pre-auricular lymphadenopathy was present L>R











Differential Diagnoses:

Bacterial conjunctivitis

- More mucopurulent discharge and papillary reaction would be expected.

Allergic conjunctivitis

- Usually bilateral and would cause a papillary response rather than follicular conjunctivitis.
- More significant itching would also be expected.

Chlamydial conjunctivitis

- A subacute onset of unilateral or bilateral redness would occur, accompanied by tearing and a watery or mucopurulent discharge.
- Follicles can be present in the inferior fornix and upper palpebral conjunctiva and there is often preauricular lymphadenopathy.
- This is usually a more chronic form of conjunctivitis and usually would be associated with a systemic chlamydial infection.

Herpes Simplex Keratitis/conjunctivitis

- Not as common a cause for viral conjunctivitis as adenoviral.
- The patient also had a recent history of URTI, which indicates that an adenoviral cause is more likely.
- No dendritic epithelial defect, no previous history

Simple Episcleritis

- This is common in young females and usually has a sudden onset of a red and uncomfortable eye within an hour.
- The redness can be variable and the attack usually peaks within 12 hours and then gradually fades over next 10-21 days.
- However, this does not cause follicular conjunctivitis.
- This is usually unilateral and would not spread from one eye to the other. Not associated with lid oedema

Uveitis

- There was no sudden onset of pain or photophobia
- Pupil reactions were normal, pupils were equal in size
- no cells or flare was found in the anterior chamber.

Toxic keratopathy

- There was no history of the eye coming into contact with any chemicals.

  Patient has used non-preserved lubricants, which is not as likely to cause this
- SPK would be more diffuse

Blepharoconjunctivitis

- More chronic condition
- Usually would not start with one eye before spreading to the other

Diagnosis: Adenoviral keratoconjunctivitis

EPIDEMIOLOGY • extremely common in the US • occurs equally in men and women • no racial predilection highly contagious—outbreaks can sometimes be traced to infected individuals or locations	SIGNS • follicular conjunctivitis (especially on the inferior palpepral conjunctiva) • watery, mucoid discharge • crusting may be evident on the lashes • edematous lids • palpable preauricular lymphadenopathy • pinpoint suconjunctival hemorrhage • in EKC, pseudomembranes and subepithelial (stromal) infiltrates can be seen • Keratitis occurs in 30% of PCF cases (c.f. 80% of EKC) and involves subepithelial infiltrates developing 1-2 weeks after the onset of conjunctivitis, representing an inflammatory reaction to the presence of viral antigens.
SYMPTOMS • "Red eye" noted by the patient • watery during the day and crusting noted in the mornings • swollen lids • patient noticed in one eye first, perhaps with later spread to the opposite eye • in pharyngoconjunctival fever—sore throat, fever, and headache may be present	TREATMENT • usually supportive • cool compress and artificial tears for comfort several times a day • prevent contagious spread (including washing sheets and pillowcases, handwashing, and cleaning of instrumentation in the physician's office). Temporary leave of absence should be considered for patients who work with the public who have active infection. • NO antibiotic or antiviral drops are routinely used. In cases where bacterial co- or super- infection is suspect, antibiotic drops may be indicated. There are no antiviral drugs approved for adenoviral conjunctivitis • in EKC only: pseudomembranes should be manually peeled every 2-3days. Topical corticosteroids may be needed (e.g. FML 0.1% q.i.d.) to prevent scarring • Betadine off-label in-office irrigation, no RCT evidence

Contraindications/Precautions:

- Steroids contraindication – HSK, fungal or bacterial keratitis

    - Those at risk of steroid response – monitor all steroid use closely
- Pregnancy category B3 – FML, category C – Flarex

Prognosis/course of disease:

- The signs and symptoms usually worsen over the first 4-7 days before self-resolution within 2 weeks.
- However, the virus continues to be shed for approximately 12 days after onset.
- The prognosis for complete resolution is excellent due to the self-limiting nature of the condition, however more severe infections may lead to the persistence of corneal infiltrates for months to years after the episode and causing glare and reduction in vision.

Note on Betadine use:

- Almost all clinical studies of Betadine demonstrated quicker resolution of disease as compared to controls receiving standard care.
- In a series of 36 patients, Abel stated that 83% reported decreased lid swelling and decreased crusting 1 day after treatment with 10% Betadine or 2 days of 5% Betadine used 3 times daily. The vast majority of patients, 94%, reported "marked improvement" of symptoms within 24 to 48 hours of treatment.
- In a prospective study of 61 patients using 2% Betadine 4 times a day for 7 days, 46% recovered completely within a week, and 77% reported "absence of general ocular discomfort" within 7 days.
- Topical povidone-iodine 0.4%/dexamethasone 0.1% ophthalmic suspension used four times daily for a minimum of five days demonstrated a decrease in disease signs, reduction in viral titre and faster resolution time in inoculated rabbit eyes.
- While protocols vary, a topical anesthetic is always instilled first because povidone-iodine causes stinging. The anesthetic is typically followed by a drop of a topical NSAID and then several drops of 5% povidone-iodine in both eyes.
- close their eyelids and move their eyes around to ensure widespread coverage. Then swab the eyelids with additional 5% povidone-iodine and, after two minutes, uses a sterile irrigating solution to thoroughly rinse the eyes. Then instill another drop of a topical NSAID and have the patient continue NSAID use for a few days.
- Note: all anecdotal so far so procedures vary as no random clinical trials have been conducted

Reference for case material and photographs:
http://www.eyerounds.org/cases/case28.htm